

Sameeran Joshi

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EDUCATION

Ph.D. in Computer Science

Aug 2022 – Present

University of Utah, Salt Lake City, UT

Advisor: Dr. Mary Hall. Graduate Research Assistant, Compilers and Tools for Outstanding Performance (CTOP) Group. Research focus: compilers and programming models for spatial dataflow hardware, hardware–software co-design, and high-performance computing.

B.E. in Computer Science

May 2015 – Apr 2019

P.V.G.'s College of Engineering and Technology (COET), Pune, India

RESEARCH INTERESTS

Programming languages, compilers, and high-performance computing, with a focus on designing new abstractions and compilation techniques for existing and emerging spatial dataflow architectures.

PUBLICATIONS

Journal Articles

- M. Hall, C. Oancea, A. C. Elster, A. Rasch, S. Joshi, A. M. Tavakkoli, R. Schulze. “Scheduling Language Chronology: Past, Present, and Future.” *ACM Transactions on Architecture and Code Optimization (TACO)*, 2025.

Conference and Workshop Papers

- A. M. Tavakkoli*, S. Joshi*, S. Singh, Y. Xu, P. Sadayappan, M. Hall. “PEAK: Generating High-Performance Schedules in MLIR.” *36th International Workshop on Languages and Compilers for Parallel Computing (LCPC)*, Oct. 2023. (*equal contribution)
- M. Hall, G. Gopalakrishnan, E. Eide, J. Cohoon, J. M. Phillips, M. Zhang, S. Y. Elhabian, A. Bhaskara, H. Dam, A. Yadrov, T. Kataria, A. M. Tavakkoli, S. Joshi, M. S. T. Karanam. “An NSF REU Site Based on Trust and Reproducibility of Intelligent Computation: Experience Report.” *EduHPC Workshop*, held at SC23.

Manuscripts in Preparation

- “Mapping a Sparse Linear Solver onto the Cerebras Wafer-Scale Engine.” Mapped a multigrid scientific application onto the Cerebras Wafer-Scale Engine’s kernel language, addressing problem layout across a million cores, inter-node dataflow, and memory mapping. Preliminary results show a 10–100× speedup over an NVIDIA GPU solver.

RESEARCH AND INDUSTRY EXPERIENCE

Compiler Intern, AIE Accelerator (NPU)

May 2025 – Aug 2025

AMD, Campbell, CA

Manager: Mahesh Ravishankar. Mentor: James Newling. Worked on a VLIW, tiled-IR compiler built on IREE and LLVM for the NPU. Supported and vectorized truncation and reduction kernels used in deep learning, achieving on-par or faster speedups across multidimensional matrix shapes.

Research Intern, Technical PhD

May 2024 – Aug 2024

Argonne National Laboratory, Lemont, IL

Supervisor: Dr. Siddhisanket Raskar. Studied code-generation challenges and opportunities for mapping HPC applications onto AI testbeds, including Cerebras, SambaNova, Groq, and GraphCore. Added GraphCore backend support to DaCe (Data-Centric Parallel Framework).

CPU Compiler Engineer, Full Time

Jun 2019 – Jun 2022

AMD, Bangalore, India

Contributed to compiler verification (unit testing, automated bug-finding tools), performance analysis tooling (built an internal tool on the LLVM BOLT project), and front-end support for Flang (Fortran, OpenMP).

Undergraduate Researcher, GCC Project

Jun 2018 – Apr 2019

Remote

Mentor: Andi Kleen (Intel); guidance from Siddhesh Poyarekar and Prathamesh Kulkarni. Extended Csmith with ~15 GNU C language extensions to fuzz the GCC compiler; found 12 critical bugs (11 fixed upstream) and raised Csmith’s GCC fuzzing coverage by 5% (line), 7% (function), 4% (branch).

HONORS AND AWARDS

- Student Travel Grant, Workshop on Sparse Tensor Computations
- Tuition Fee Waiver Scheme (TFWS) Scholarship — 90% of tuition waived, undergraduate studies

PROFESSIONAL SERVICE

- Artifact Evaluation Committee: CGO 2025, ASPLOS 2025
- Program Committee Member: LLVM Developers' Meeting 2021
- Volunteer: SC25, CppCon 2021, CppOnSea 2021

TALKS AND OUTREACH

- "Extending Csmith with GCC C-Language Extensions." Pune Kernel Meetup, Pune, India.
- Guest talk on open-source contribution opportunities for undergraduates. P.V.G.'s COET, Pune, India.